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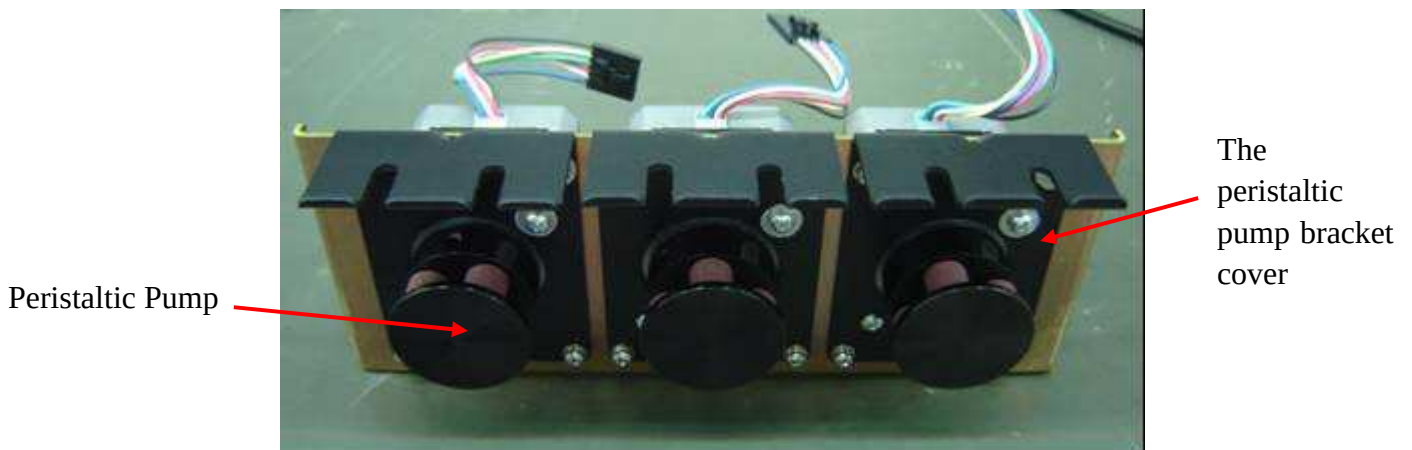
1. Installation of ISE assembly

1.1 Install peristaltic pump

Parts needed

PN	Part Description	Qty.
M04-021000---	Flat washer - GB/T97.1-20022.5	12
M04-021002---	Standard spring washer GB/T93-1987 2.5	12
M04-051045---	Philips small head screw GB/T823-1988 M2.5X6	12
BA40-20-61457	The peristaltic pump bracket cover	1

Use 12 M2.5x6 screws to fasten 3 peristaltic pumps to the bracket. Pay attention to the cable of the pump. Don't scratch it.



Picture 1

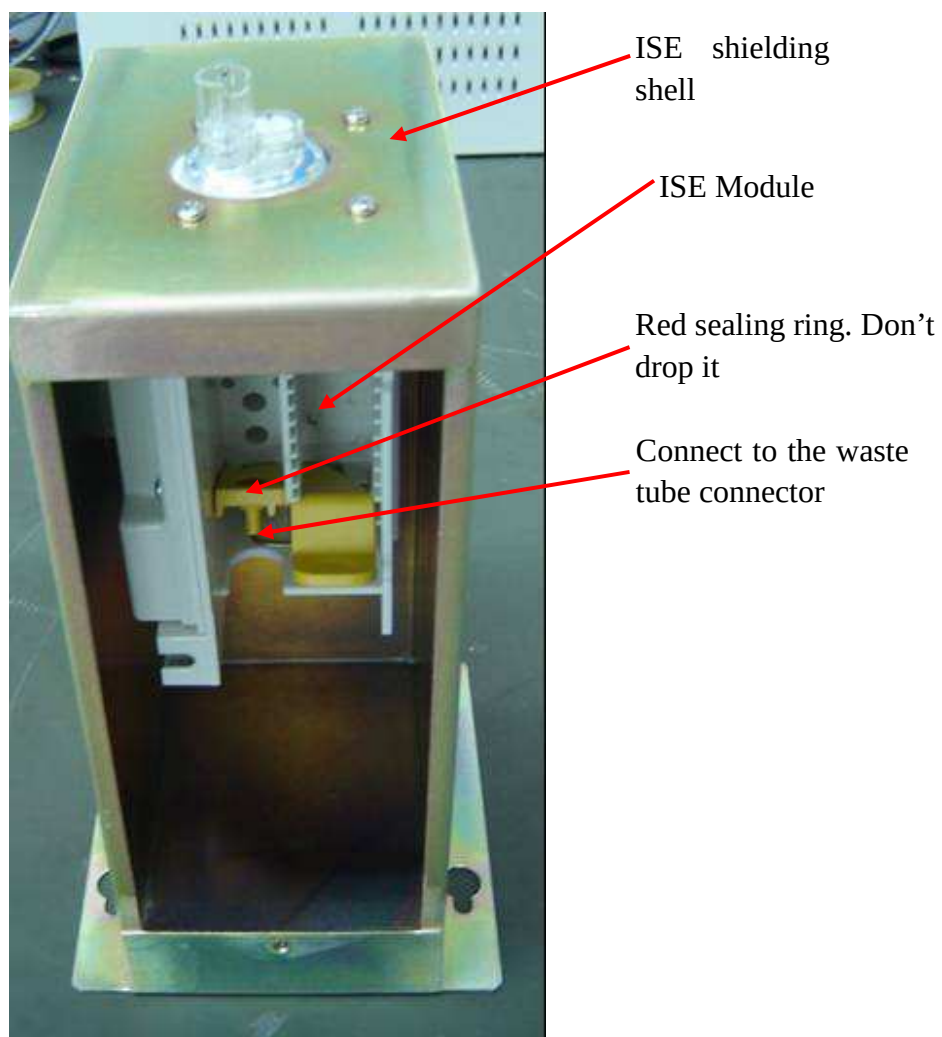
1.2 Install ISE Module

Parts needed

PN	Part Description	Qty
BA34-10-63674	ISE Module	1
M04-051125---	Stainless steel cross screw #6-32X5/16 UNC-2A	4
BA40-20-61455	ISE shielding shell	1

Use 4 cross screws to fasten ISE Module to ISE shielding shell. Pay attention don't drop the red

sealing ring.



Picture 2

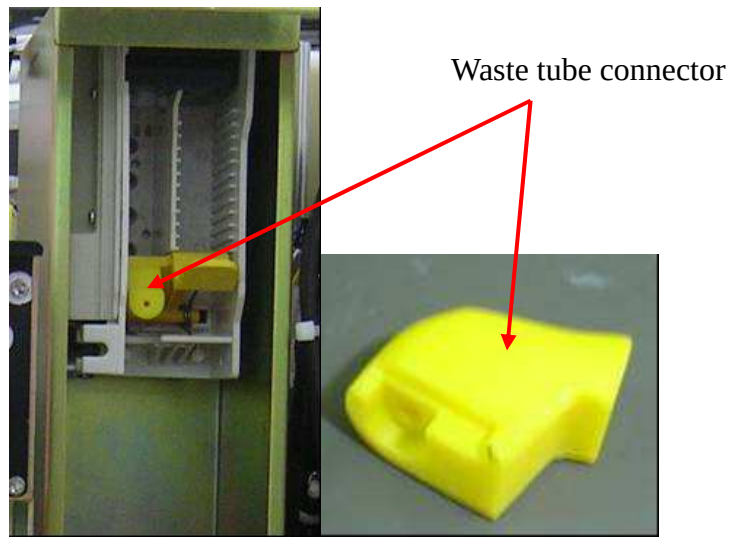
1.3 Installation of waste tube connector

Parts needed

PN	Part Description	Qty
BA34-10-63666	ISE Module waste tube connector	1
M90-100059---	Ribbon	45-47mm

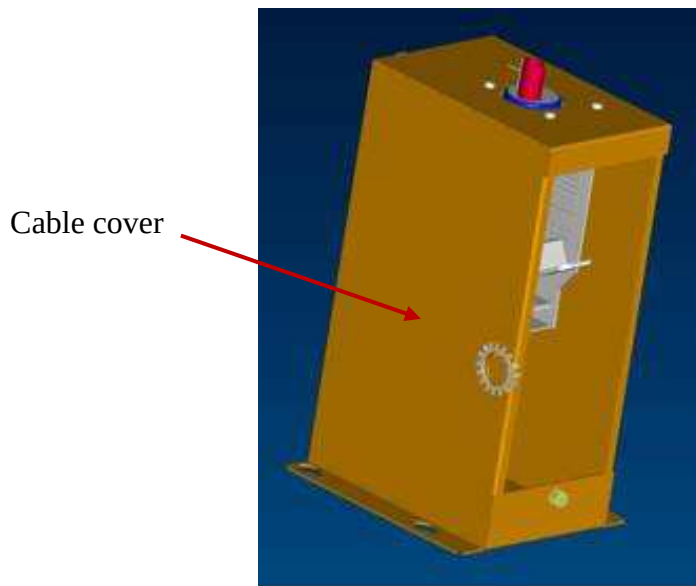
1. Connect waste tube connector to ISE module. Pay attention to the direction of the connector

and make sure put it to the place. No leakage.



Picture 3

2. Cut a cable cover with the length of around 45 to 47 mm. Put it into the ISE shielding shell hole



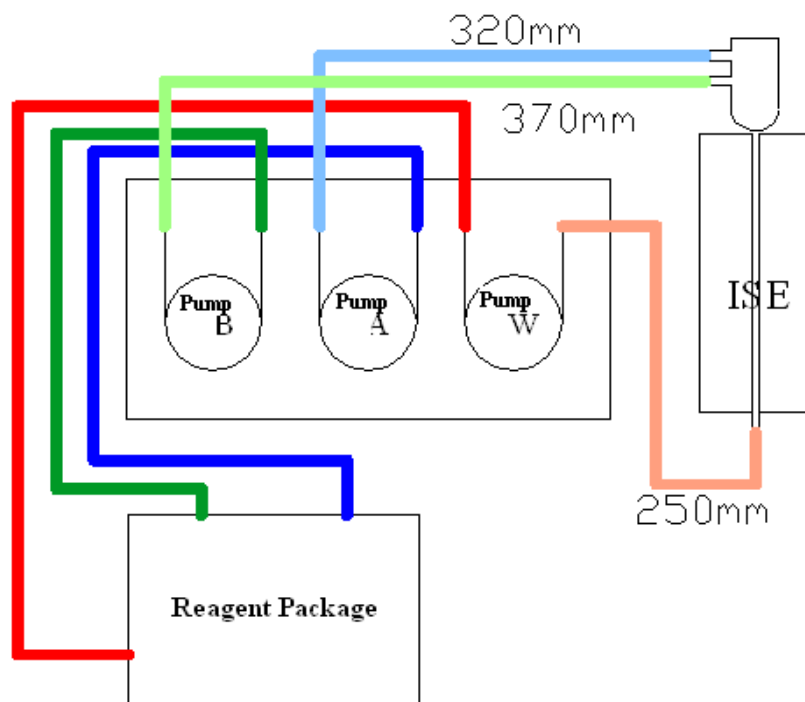
Picture 4

1.4 Tube and cable connection

Parts needed

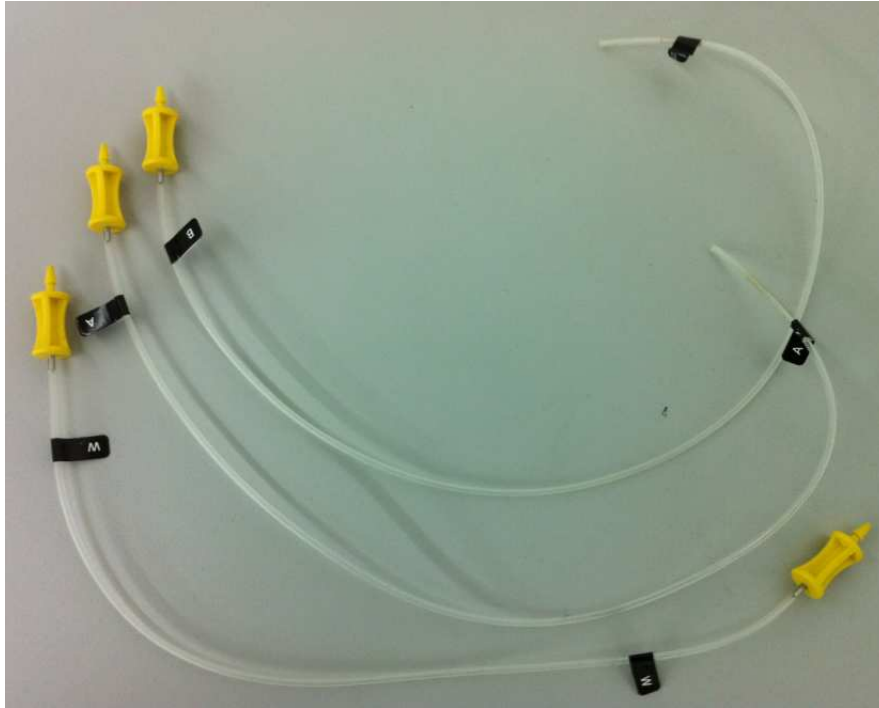
PN	Part Description	Qty
801-BA34-00106-00	ISE Fluid Tubing Kit	
BA40-20-73066	ISE Module Patch Cord	1
BA34-10-63667	Tubing Adapter	4

1. Cut out ISE liquid tube with specified length (Tube A: 320mm; Tube B: 370mm; Tube W: 250mm)。



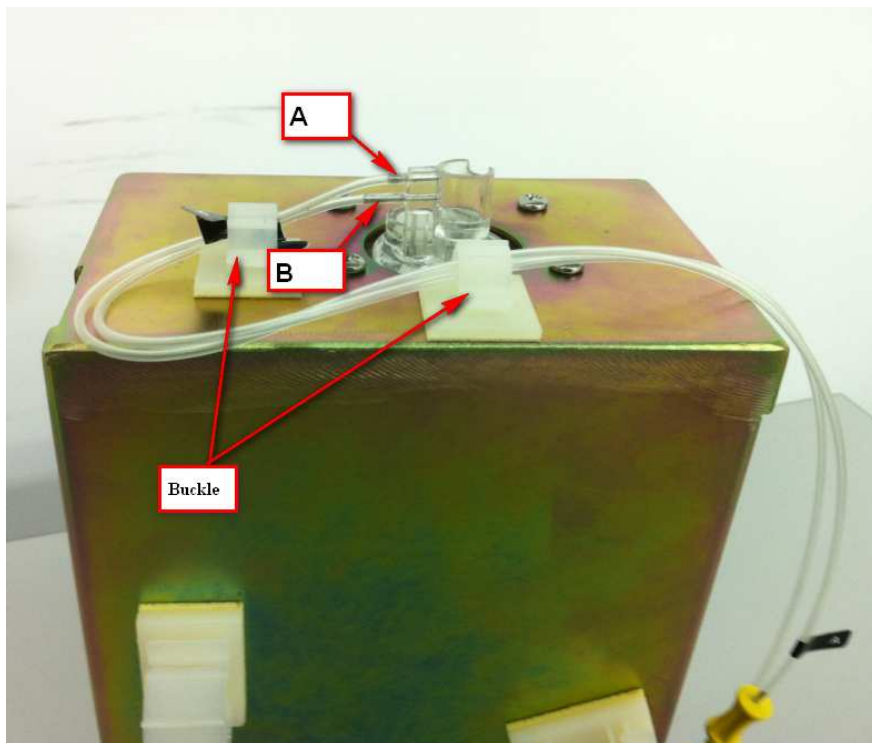
Picture 5

2. Connect the tube (W/A/B) to the tubing adaptor as the picture below. The length of the tube that insert in the adaptor should not less than 3.5mm.



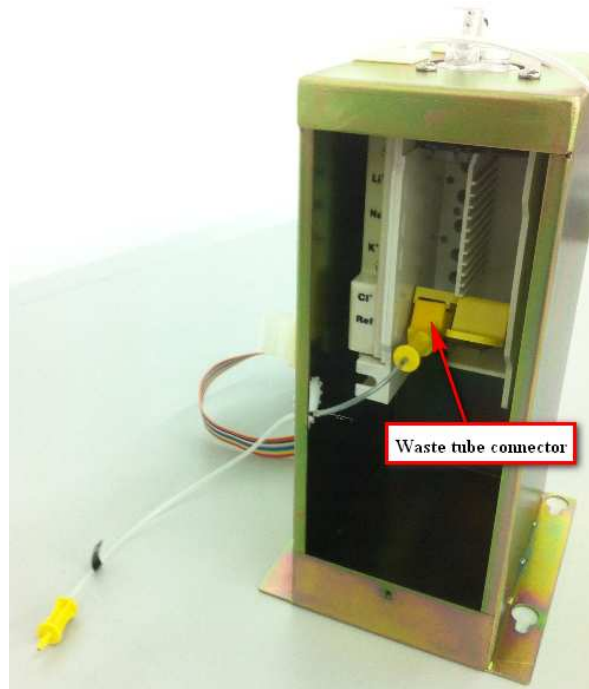
Picture 6

3. Connect tube A and B to the ISE module as picture below. Tube A is above, tube B is below. Penetrate the tube A and B through the buckle. Make sure the tubes are not bent.



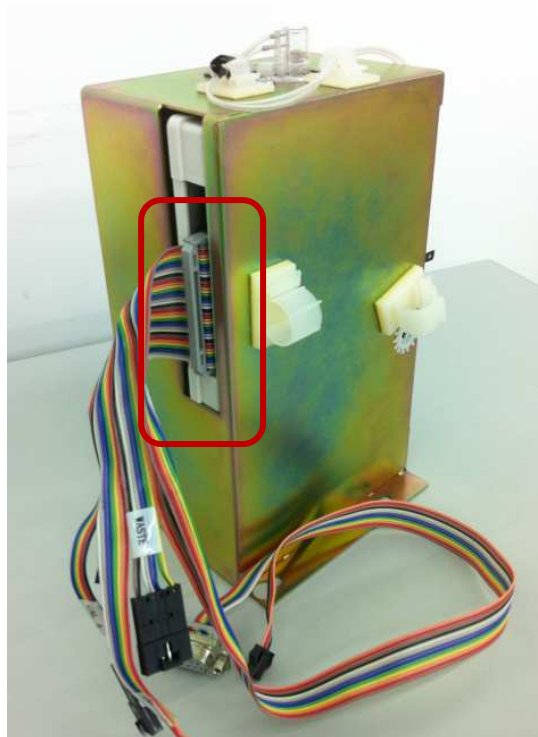
Picture 7

4. Connect tubing adaptor of tube w to waste tube connector and put the other adaptor through the hole of shielding shell as the picture below.



Picture 8

5. Connect ISE Module Patch Cord to ISE main control board.

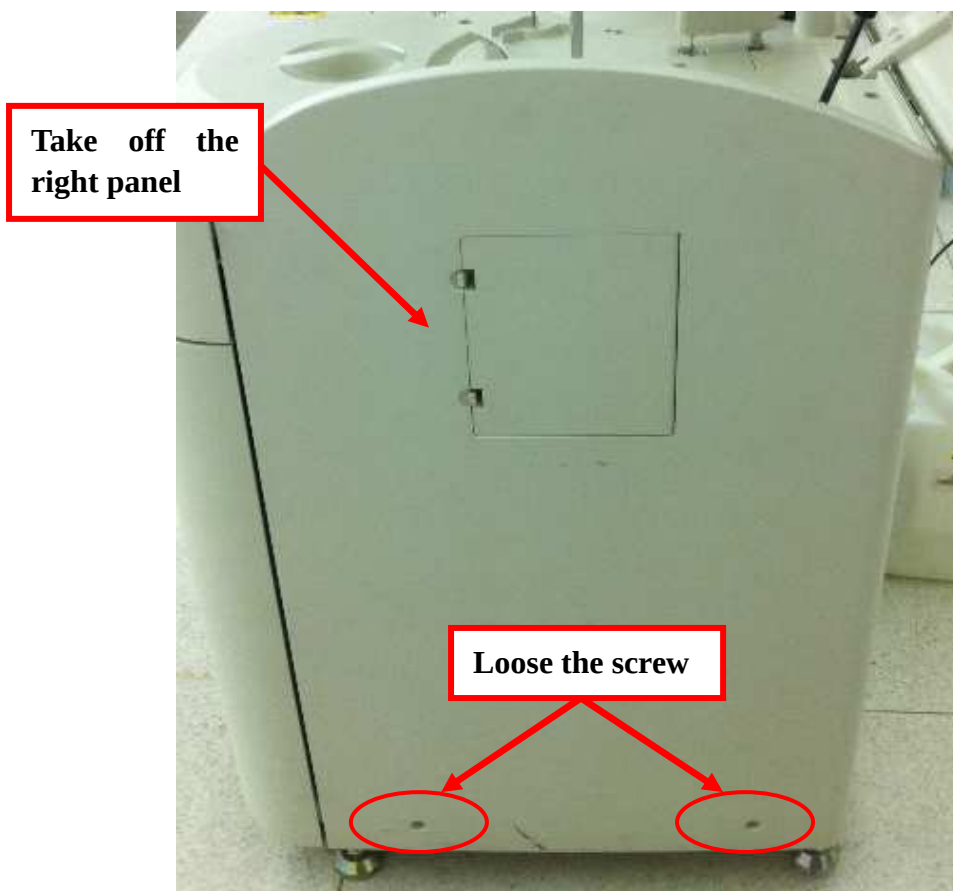


Picture 9

2. Install ISE module to main unit

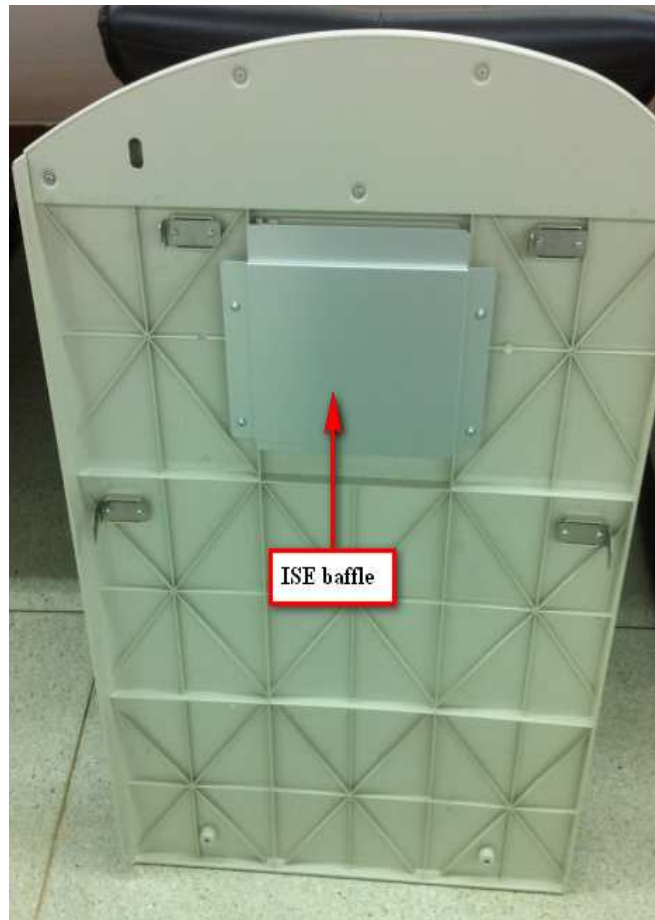
2.1 Take off the panel

1. Take off the right panel of the main unit



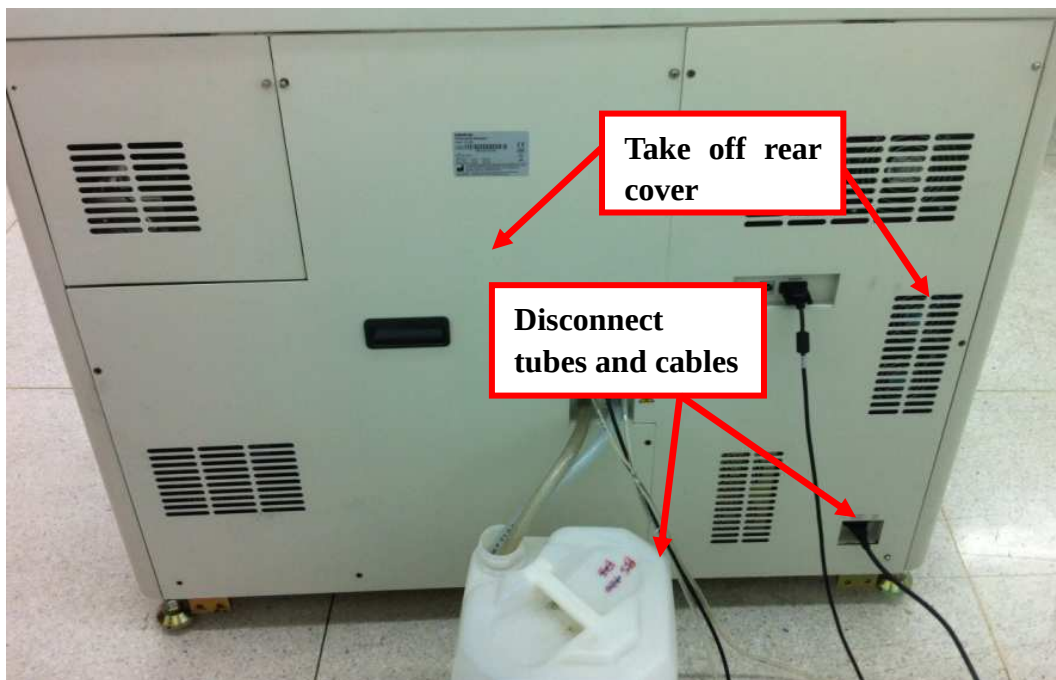
Picture 10

2. Take off the ISE baffle



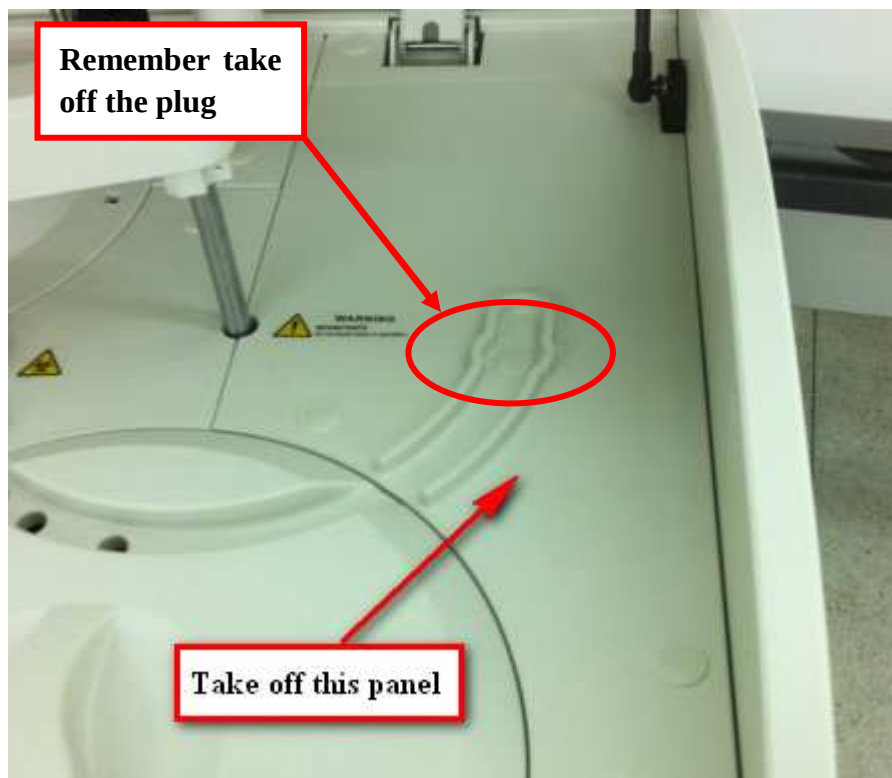
Picture 11

3. Take off the rear cover and disconnect the tubes and cables.



Picture 12

4. Take off the panel as the picture shows



Picture 13

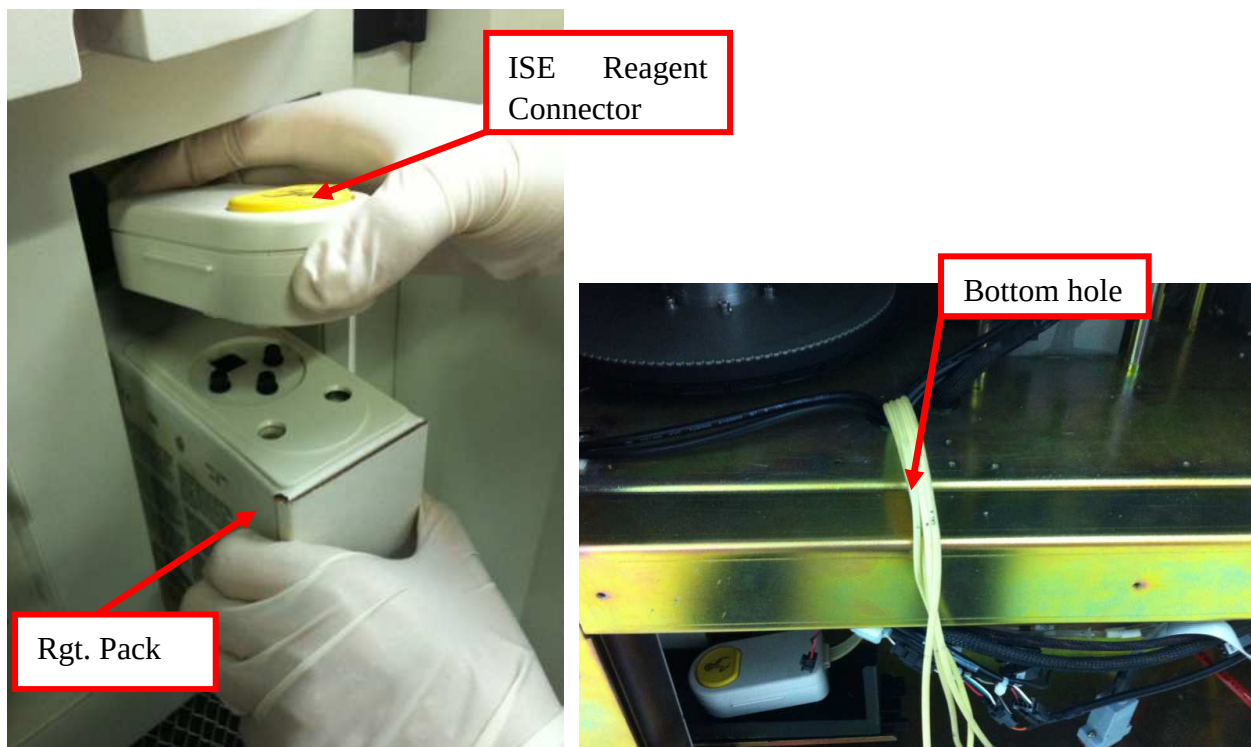
2.2 Install ISE Module

Fasten ISE module and peristaltic pump

Parts needed

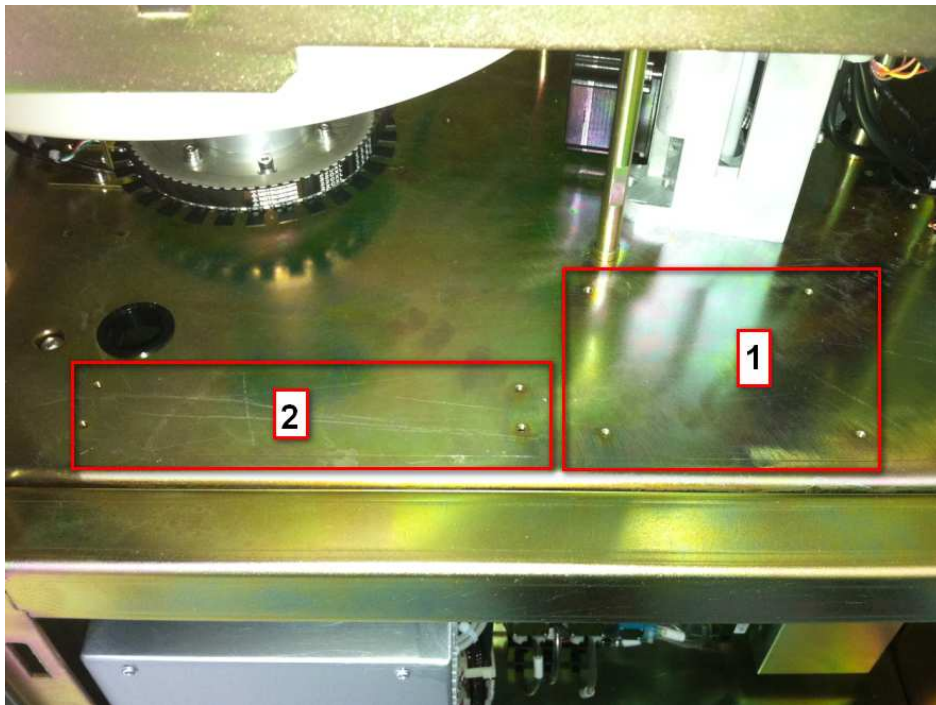
PN	Part Description	Qty.
BA34-10-63812	ISE Reagent Connector	1
M04-051141---	Cross Screw	8

1. Put the Rgt. Pack into the bracket. Put the tubes through the hole the right picture shows.



Picture 14

2. Use M4*8 cross screws fasten ISE module on position 1 peristaltic pump on position 2.

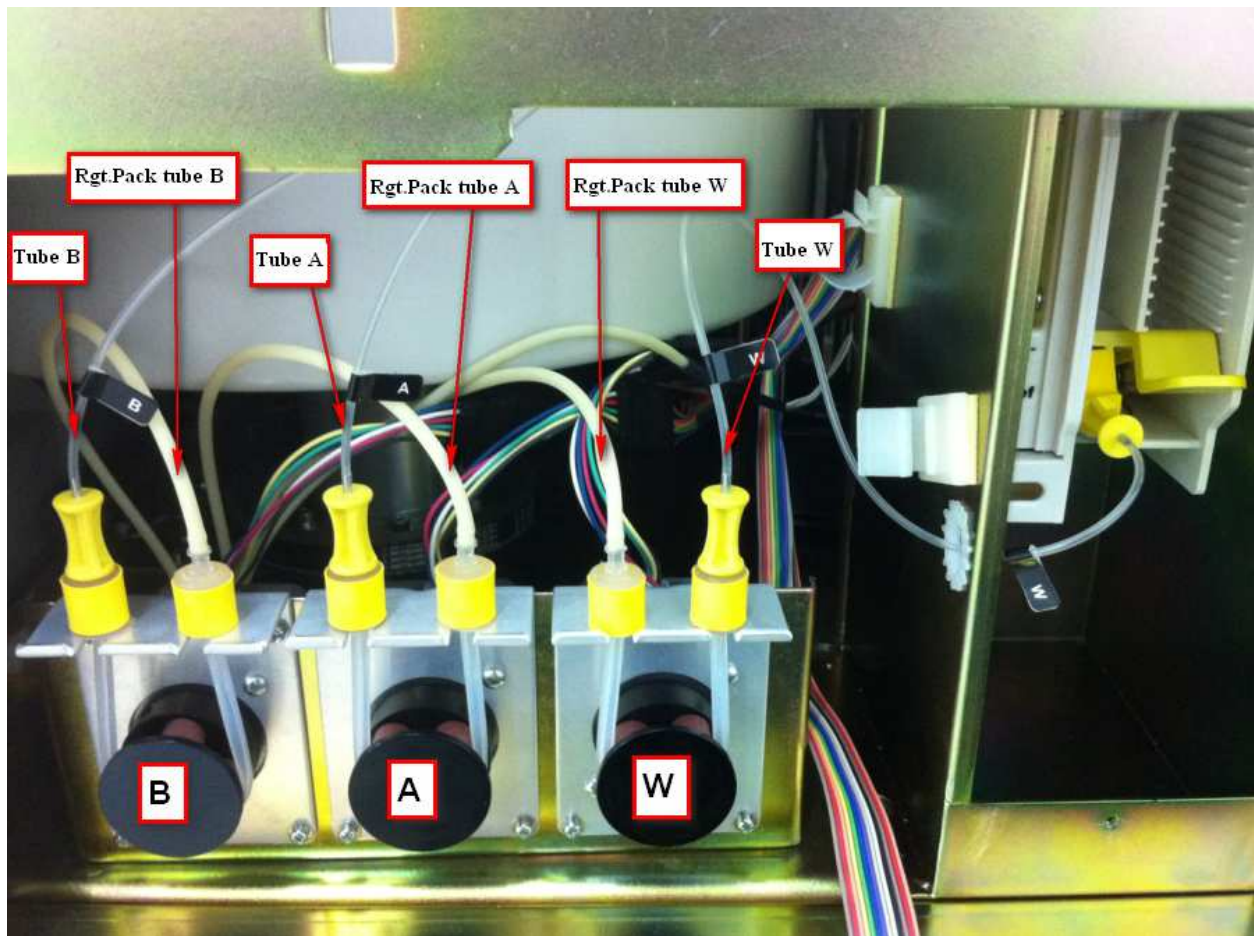


Picture 15



Picture 16

3. Connect the tubes of the pumps as the picture below. Connect the cables according to the mark on ISE Module Patch Cord.



Picture 17

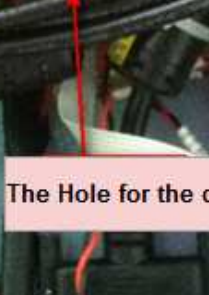
Connect ISE module communication cable and power cable

Parts needed

PN	Part Description	Qty.
BA40-20-73065	ISE power cable	1
BA40-20-61591	Connection Wire for Main Control Board and ISE Module	1

-

Picture 18

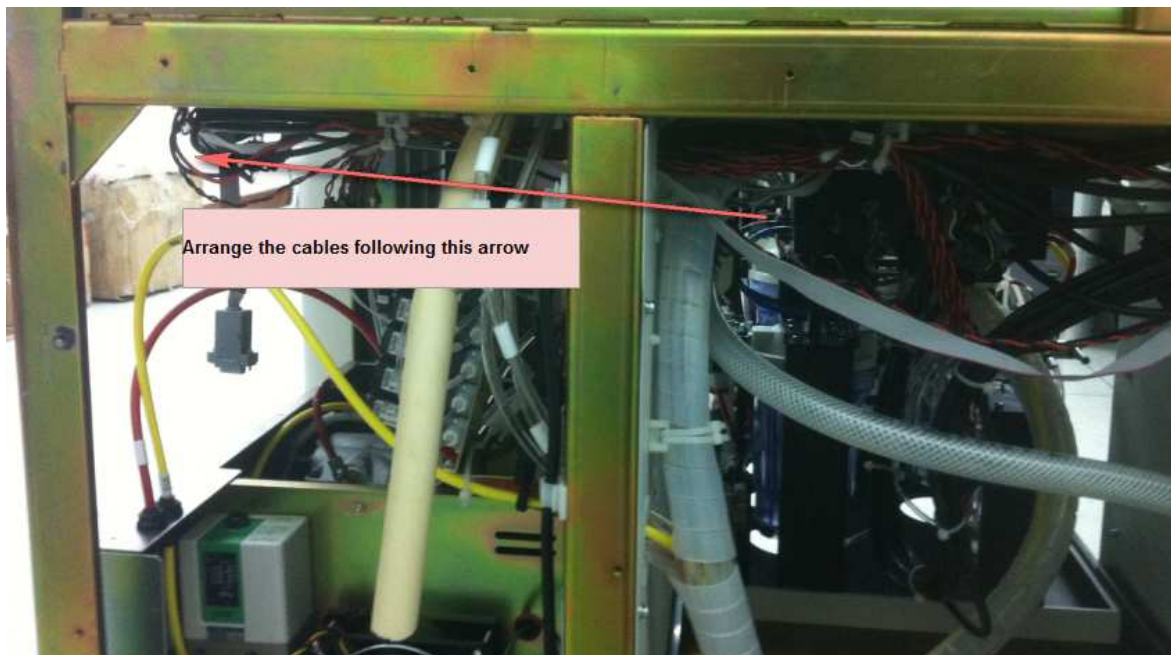
- 
- The Hole for the cables
- J4

Picture 19

3. On the rear of the machine, arrange these two cables to the left of the machine near the ISE module along the direction of the arrow below, and fix them with the other cables by the ribbons.



Picture 20



Picture 21

4. Connect the ISE module cable with the ISE module power cable and the other cables of the ISE module.



Picture 22

3. Alignment of the ISE module

3.1 Install the electrodes and the reagent package

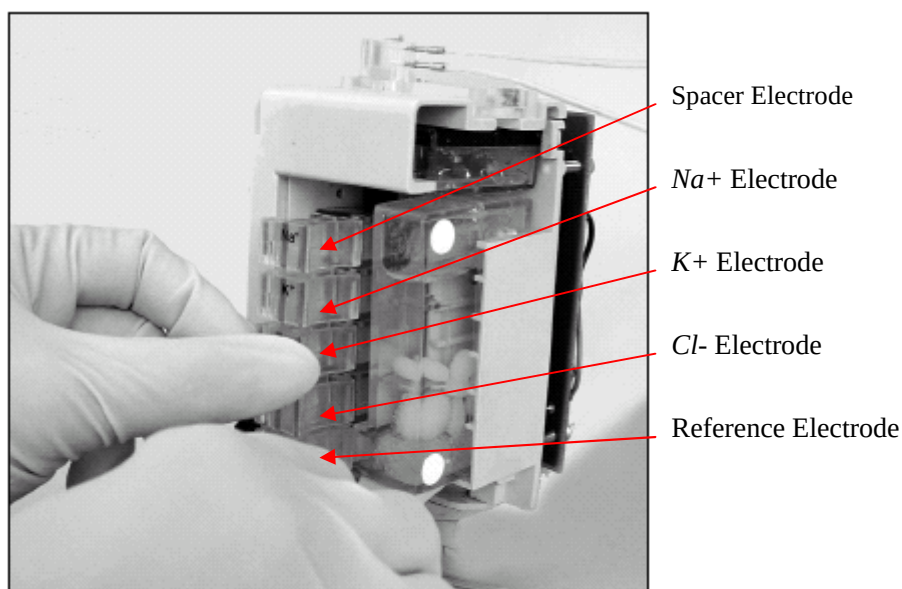
Parts needed

PN	Part Description	Qty.
801-BA10-00031-00	Chloride Electrode	1
801-BA10-00033-00	Potassium Electrode	1
801-BA10-00034-00	Sodium Electrode	1
801-BA34-00067-00	MR Dummy Electrode	1
BA34-10-63643	Spacer Electrode	1
BA34-30-63681	ISE reagent package	1

Install the reference, Cl⁻, K⁺, Na⁺ and the spacer into the ISE module from the bottom to the top.

Caution:

1. Please be careful not to lose the O-ring seals;
2. It is recommended to install the electrodes from the bottom to the top. You need to press down the base when you install the last electrode, to make sure the electrode is fixed well.
3. Double check if the electrodes are fixed well.



Picture 23

Please check as the picture below after finish the installation.



Picture 24

Connect the reagent package and the ISE Module Wand as below, and put them on the reagent package support.



Picture 25

3.2 Adjust the position of the sample probe

1. Adjust the horizontal position of the sample probe.

Target: The tip of the probe should be 1mm around the center of the ISE cup.

- 1) Make sure the ISE module is installed, the operation of the jumper of the mainboard has been done.
- 2) Choose “Utilities→Maintenance→Alignment→Sample Probe Unit →Sample Probe Rotary to ISE” to do the alignment.

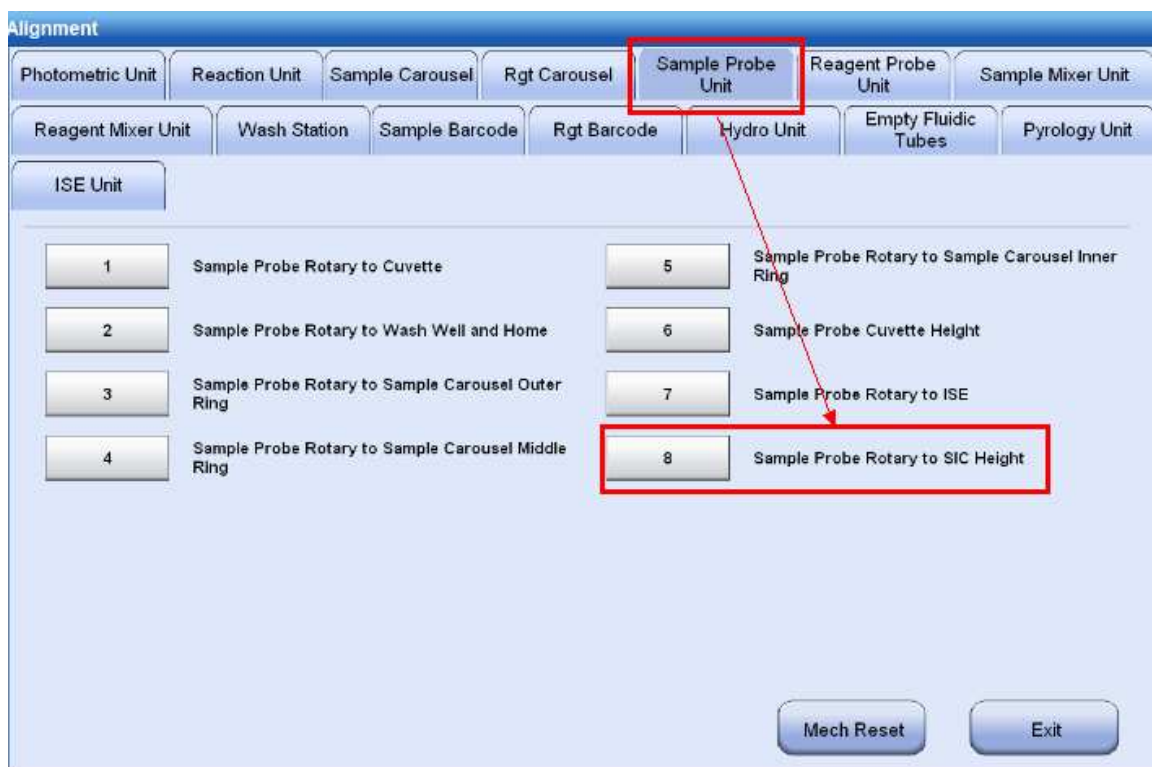


- 3) Click “down” arrow to make the probe down to the above of ISE module. Click “left” and “right” arrow to adjust the horizontal position of the sample probe. The tip of the probe should be 1mm around the center of the ISE cup.
- 4) If adjusting the parameter cannot make the probe to the required position, please loose the screw that fasten the ISE module on the big raft and adjust the position of ISE module.
- 5) Confirm the position
- 6) Click “continue” to save the parameter and exit.

2. Adjust the vertical position of the sample probe

Target: The tip should be aligned with the transition line.

- 1) Confirm that the alignment of horizontal position of the sample probe is done.



- 2) According to the guidance of the software, click “up and down” arrows to make the tip of the probe aligned with the transition line as the picture shows.
- 3) Confirm the position. Click “continue” to save the parameter and exit

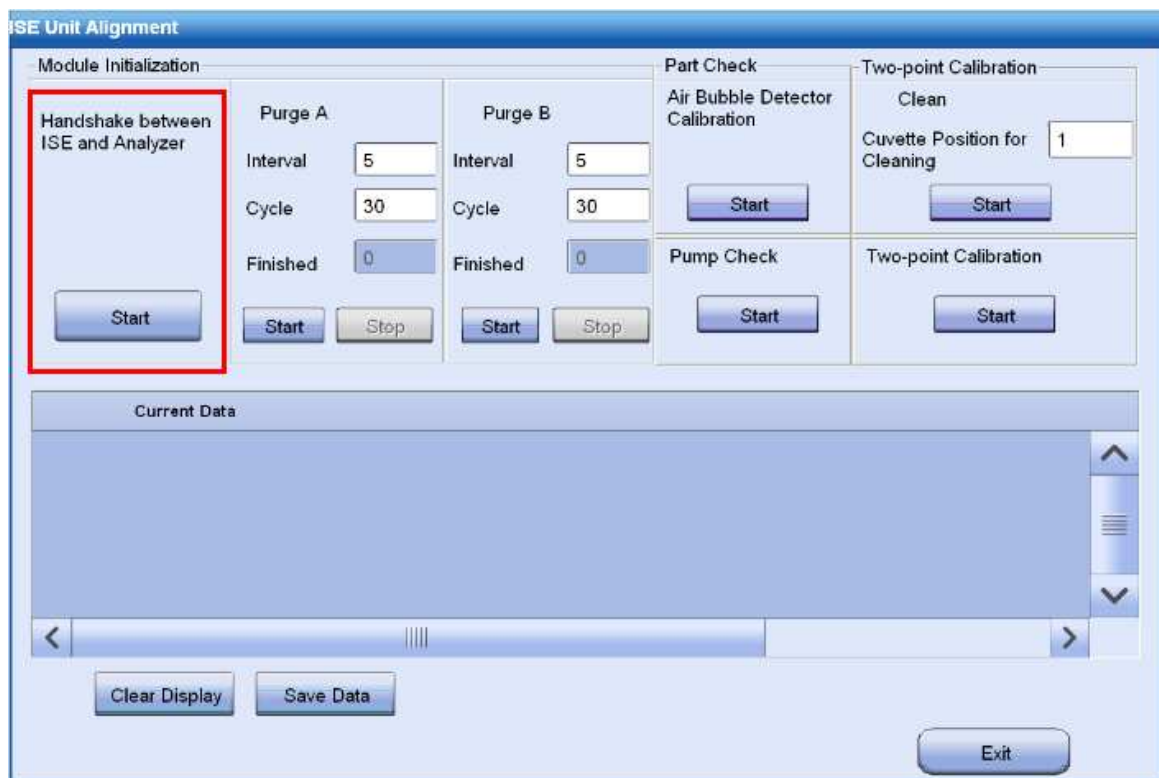


Picture 26

3.3 ISE Functional Module Alignment

Initialization

Click “Utilities→Maintenance→Alignment→ISE Unit go to ISE Unit test.

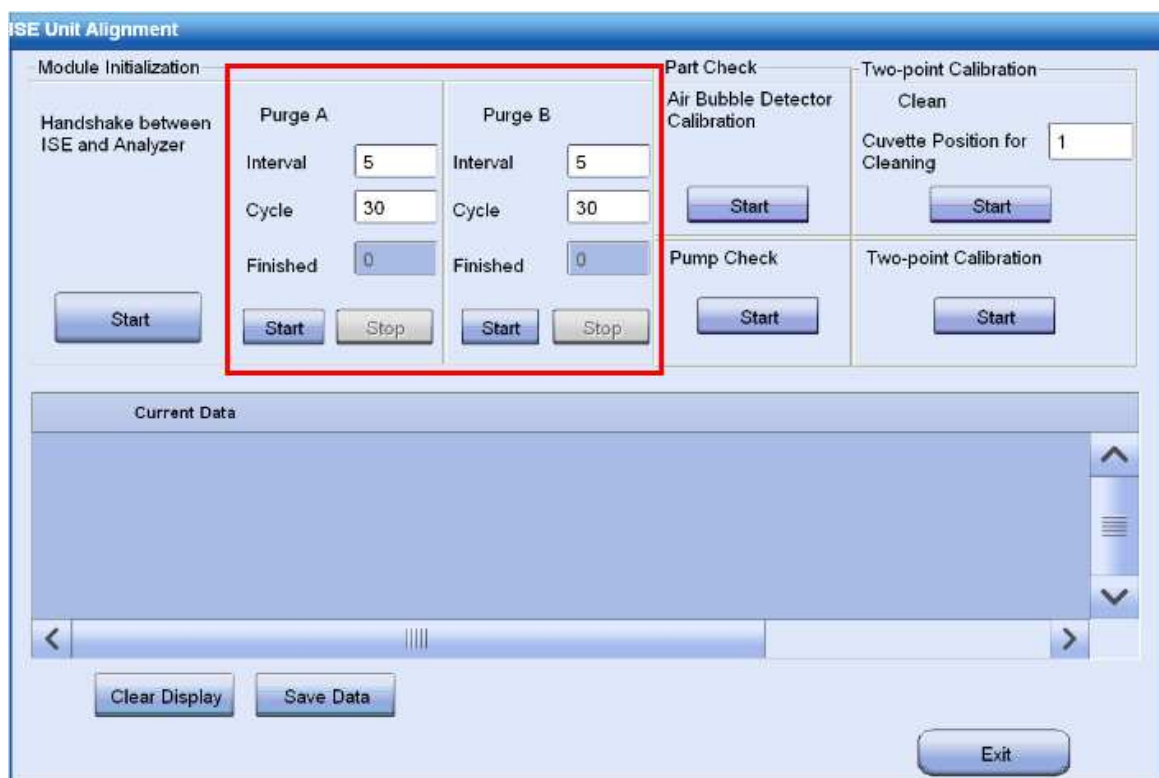


Picture 27

Click “Start” to initialize the ISE module, when software send back the data “<ISE!>” in the window “Received” data and the next information is handshake successful it means the communication of ISE module is OK.

Purge A and Purge B Execution

Click Purge A button, when <ISE!> occurs in the Received Data Section, it means Solution A has been purged one process. During this process, Pump A and Waste Pump will be working and solution cycling out. Repeat the step, and until the tube fully filled with the solution. Also you could input value into Interval (5s is appreciated) and Times to make the Purge Process done automatically. If you found no solution cycling in the tube or the speed for filling is too slow, please check tube connection and pump working status. Purge B process is the same with Purge A.

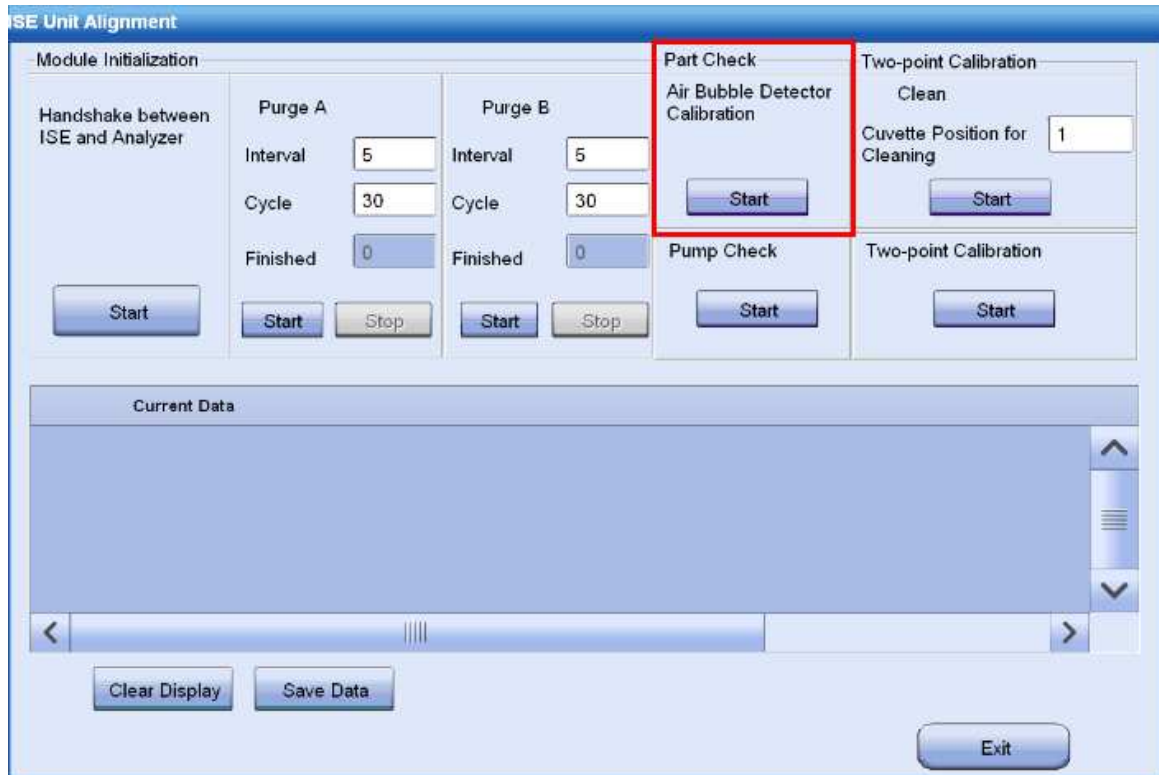


Picture 28

Note: During the Purge Process, please check the pump working status. All the three pumps should rotate clockwise without leakage.

Bubble Calibrate

Click Bubble Calibrate button and you will receive the data whose format should be like <BBC A xxx M xxx L xxx> . If you got some errors about the bubble, you need repeat the step to calibrate again. If the warning message cannot be reduced, you have to replace the whole ISE Module.

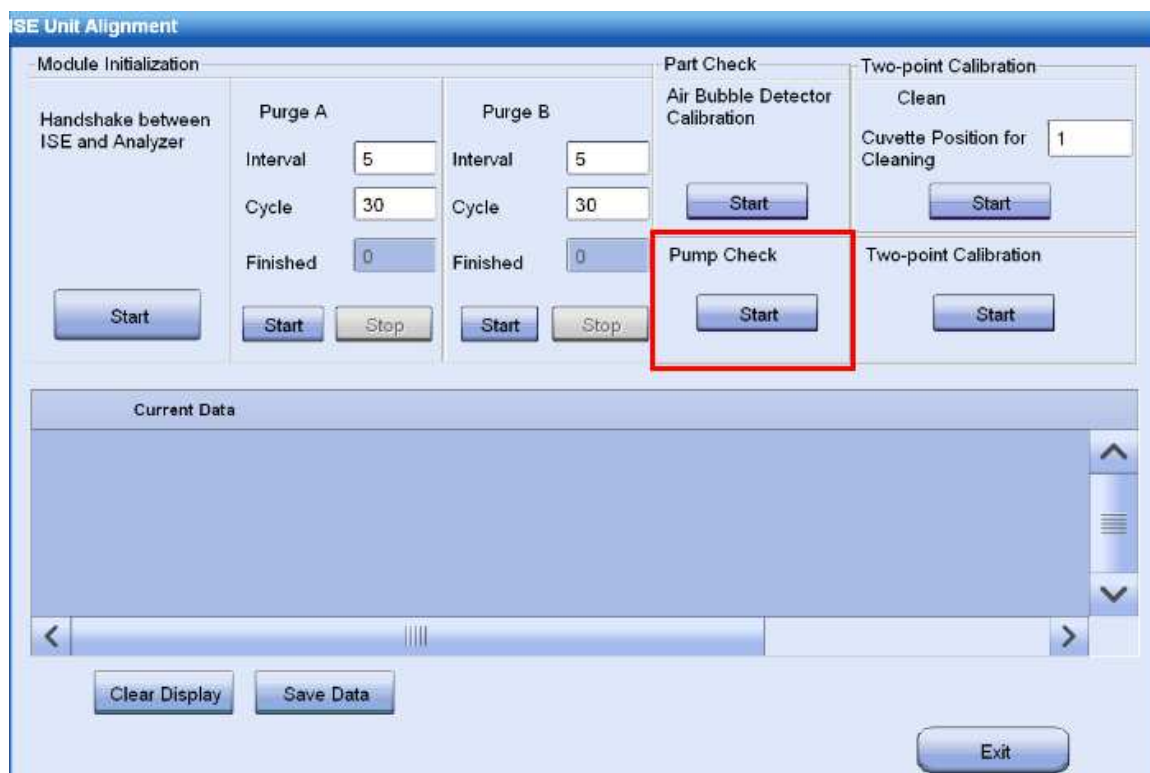


Picture 29

Calibrate Pump

Click Calibrate Pump, you will receive the data whose format is like <PMC A xxxx B xxxx W xxxx>, and the A, B, W value should be between 1500~3000. If you repeat the step some times and still occur the warning message about pump, you need to check the pump status and tubing connection. Or replace the pump and tubing when needed.

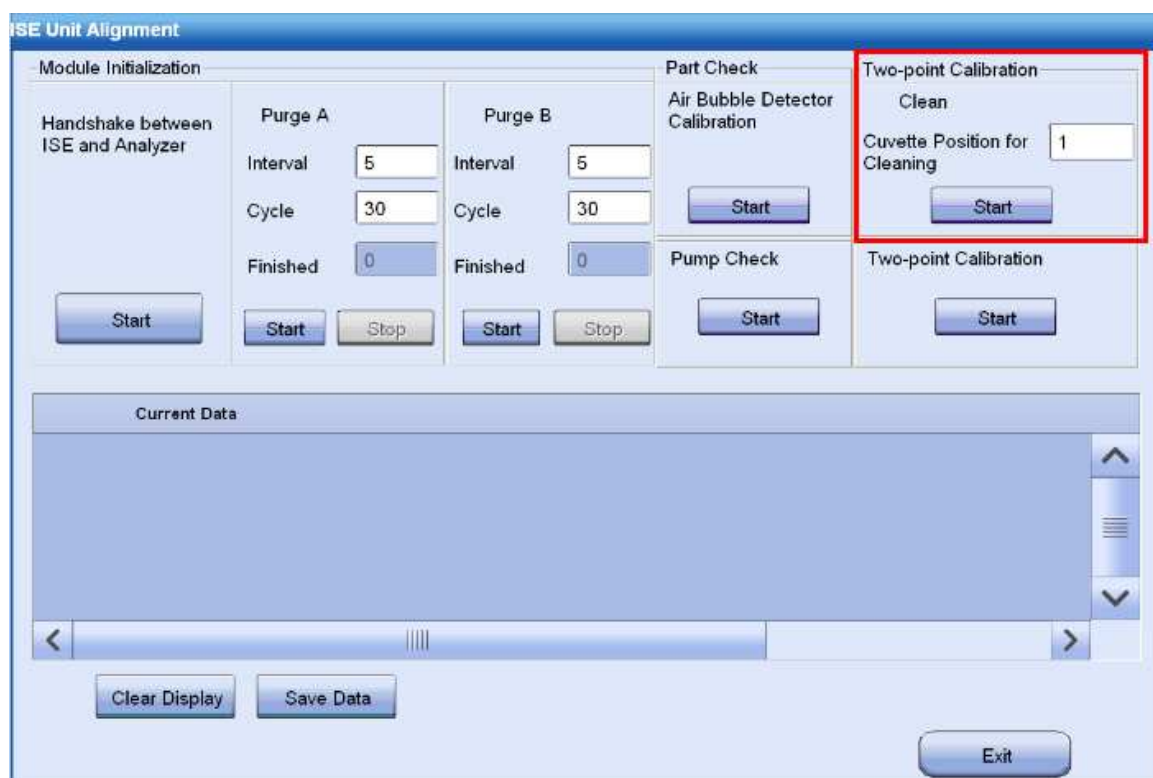
Note: When operate the step, sample probe will join the calibration, so this step should after the Probe Alignment.



Picture 30

Wash

Please set the Cuvette Position and then fill the enough wash solution into the cuvette, 100ul solution will be consumed per time. After click the Wash Button and you got the <ISE!> in the Received Data, it means you have done successfully.

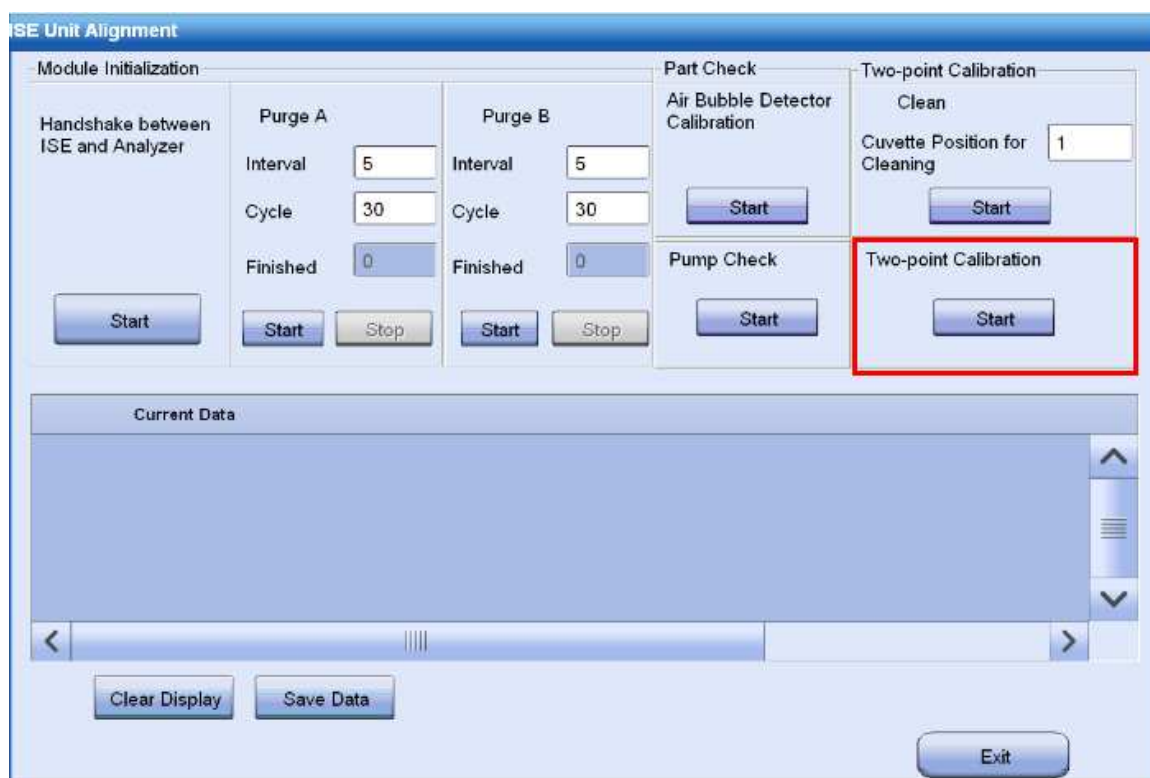


Picture 31

Calibrate

We should wait 15min after Wash. Then click Calibrate and module will do the calibration automatically. About 70s later, we will get the Calibration data whose format is like <Li xx.xx Na xx.xx K xx.xx Cl xx.xx xxxxxxxx> .

If you got some error messages during the calibration, you need repeat calibrating and if the warning message could not be eliminated, you need replace the corresponding electrode.



Picture 32

4. Recover the panel

Put away the panels including the ISE shielding shell.



Picture 33

5. Attention

1. Don't turn off the main switch if the machine has ISE module. Because the machine primes every half an hour automatically. Long time no priming can lead to damage of electrode
2. After the upgrade of ISE module. We suggest standing and auto-prime for 24 hours before calibration and normal test.
3. After the ISE cleaning, please wait at least 15 minutes before calibration because the electrode is unstable right after the cleaning.
4. When execute calibration, we suggest calibrating 3-4 times, waiting the calibration data to be stable before control or sample test.